

JEE MAINS & IIT JEE ADVANCED

MATHEMATICS

SEQUENCE & SERIES

DIBY (DO IT BY YOURSELF)

By: Sachin Jakhar (Defender Mamu)



DIBY-01

Let a_n be the n^{th} term of an A.P. If $S_n = a_1 + a_2 + a_3 + \cdots + a_n = 700$, $a_6 = 7$ and $S_7 = 7$, then a_n is equal to: [JEE Mains-2025]

- (A) 56 (B) 65 (C) 64 (D) 70

[Ans.: C]

DIBY-02

If the first term of an A.P. is 3 and the sum of its first four terms is equal to one-fifth of the sum of the next four terms, then the sum of the first 20 terms is equal to

[JEE Mains-2025]

- (A) -120 (B) -1200 (C) -1020 (D) -1080

[Ans.: D]

DIBY-03

The roots of the quadratic $3x^2 - px + q = 0$ are 10^{th} and 11^{th} terms of an arithmetic progression with common difference $\frac{3}{2}$. If the sum of the first 11 terms of this arithmetic progression is 88, then $q - 2p$ is equal to [JEE Mains-2025]

[Ans.: 474]

DIBY-04

In an arithmetic progression, if $S_{40} = 1030$ and $S_{12} = 57$, then $S_{30} - S_{10}$ is equal to :

[JEE Mains-2025]

- (A) 505 (B) 525 (C) 510 (D) 515

[Ans. : D]

DIBY-05

If $a_1, a_2, a_3, \dots$ are in A.P. such that $a_1 + a_7 + a_{16} = 40$, then the sum of the first 15 terms of this A.P. is

[JEE Mains-2019]

- (A) 200 (B) 280 (C) 120

[Ans. : A]

DIBY-06

If 19th terms of non-zero A.P. is zero, then its (49th term) : (29th term) is:

[JEE Mains-2019]

- (A) 4 : 1 (b) 1 : 3 (c) 3 : 1 (D) 2 : 1

[Ans. : c]

DIBY-07

Let S_n be the sum of the first n terms of an arithmetic progression. If $S_{3n} = 3S_{2n}$, then the value of $\frac{S_{4n}}{S_{2n}}$ is: [JEE Mains-2021]

- (A) 2 (B) 6 (C) 8 (D) 4

[Ans. : B]

DIBY-08

Let $a_1, a_2, a_3, \dots$ be an A.P. If $\frac{a_1 + a_2 + \dots + a_{10}}{a_1 + a_2 + \dots + a_p} = \frac{100}{p^2}$, $p \neq 10$, then $\frac{a_{11}}{a_{10}}$ is equal to: [JEE Mains-2021]

- (A) $\frac{21}{19}$ (B) $\frac{100}{121}$ (C) $\frac{121}{100}$ (D) $\frac{19}{21}$

[Ans. : A]



DIBY-09

If $\log_3 2, \log_3(2^x - 5), \log_3\left(2^x - \frac{7}{2}\right)$ are in an arithmetic progression, then the value of x is
equal to [JEE Mains-2021]

[Ans. : 3]

DIBY-10

If the 10^{th} term of an A.P. is $\frac{1}{20}$ and its 20^{th} term is $\frac{1}{10}$, then the sum of its first 200 terms is
[JEE Mains-2020]

- (A) 100 (B) $50\frac{1}{4}$ (C) $100\frac{1}{2}$ (D) 50

[Ans. : C]

DIBY-11

If the sum of the series $20 + 19\frac{3}{5} + 19\frac{1}{5} + 18\frac{4}{5} + \dots$ upto n^{th} term is 488 and the n^{th} term is negative, then **[JEE Mains-2020]**

(A) $n = 41$

(B) n^{th} term is $-4\frac{2}{5}$

(C) $n = 60$

(D) n^{th} term is -4

[Ans. : D]

DIBY-12

Five numbers are in A.P., whose sum is 25 and product is 2520. If one of these five numbers is $\frac{-1}{2}$, then the greatest number amongst them is **[JEE Mains-2020]**

(a) $\frac{21}{2}$

(B) 16

(c) 27

(D) 7

[Ans. : B]

DIBY-13

If $3^{2\sin 2\alpha-1}$, 14 and $3^{4-2\sin 2\alpha}$ are the first three terms of an A.P. for some α , then the sixth term of this A.P. is: **[JEE Mains-2020]**

- (A) 65 (B) 78 (C) 81 (D) 66

[Ans.: D]

DIBY-14

If a_1, a_2, a_3 and b_1, b_2, b_3 are in A.P., and $a_1 = 2, a_{10} = 3, a_1 b_1 = 1 = a_{10} b_{10}$, then $a_4 b_4$ is equal to. **[JEE Mains-2022]**

- (A) $\frac{35}{27}$ (B) 1 (C) $\frac{27}{28}$ (D) $\frac{28}{27}$

[Ans.: D]

DIBY-15

A software company sets up m number of computer systems to finish an assignment in 17 days. If 4 computer systems crashed on the start of the second day, 4 more computer systems crashed on the start of the third day and so on, then it took 8 more days to finish the assignment. The value of m is equal to: **[JEE Mains-2024]**

- (A) 125 (B) 150 (C) 180 (D) 160

[Ans.: B]

DIBY-16

Consider two sets A and B , each containing three numbers in A.P. Let the sum and the product of the elements of A be 36 and p respectively and the sum and the product of the elements of B be 36 and q respectively. Let d and D be the common differences of AP's in A and B respectively such that $D = d + 3, d > 0$. If $\frac{p+q}{p-q} = \frac{19}{5}$, then $p - q$ is equal to

[JEE Mains-2025]

- (A) 600 (B) 450 (C) 630 (D) 540

[Ans. : D]

DIBY-17

Consider an A. P. of positive integers, whose sum of the first three terms is 54 and the sum of the first twenty terms lies between 1600 and 1800. Then its 11th term is:

[JEE Mains-2025]

- (A) 84 (B) 90 (C) 108 (D) 122

[Ans. : B]

DIBY-18

Number of 4 digit integers less than 2800 which are either divisible by 3 or by 11 is equal to ?
[JEE Mains-2023]

[Ans. : 710]

DIBY-19

If where n is an even integer, is an arithmetic progression with common difference 1, and $\sum_{i=1}^n a_i = 192, \sum_{i=1}^{n/2} a_{2i} = 120$, then n is equal to [JEE Mains-2022]

- (A) 48 (B) 96 (C) 92 (D) 104

[Ans. : B]

DIBY-20

Let S_1 be the sum of first $2n$ term of an arithmetic progression. Let S_2 be the sum of first $4n$ terms of the same arithmetic progression. If $(S_2 - S_1)$ is 1000, then the sum of the first $6n$ terms of the arithmetic progression is equal to [JEE Mains-2021]

- (A) 5000 (B) 1000 (C) 7000 (D) 3000

[Ans. : D]



DIBY-21

The sum of all natural numbers ' n ' such that $100 < n < 200$ and H.C.F. $(91, n) > 1$ is

[JEE Mains-2019]

- (A) 3221 (B) 3121 (C) 3203 (D) 3303

[Ans. : B]

DIBY-22

The series of positive multiples of 3 is divided into sets : $\{3\}$, $\{6, 9, 12\}$, $\{15, 18, 21, 24, 27\}$,
..... Then the sum of the elements in the 11th set is equal to

[JEE Mains-2022]

[Ans. : 6993]

DIBY-23

Let $a_1, a_2, \dots, a_{30}$ be an A.P., $S = \sum_{i=1}^{30} a_i$ and $T = \sum_{i=1}^{15} a_{2i-1}$. If $a_5 = 27$ and $S - 2T = 75$, then a_{10} is equal to

[JEE Mains-2019]

- (A) 52 (B) 57 (C) 47 (D) 42

[Ans. : A]

DIBY-24

Let $a_1, a_2, \dots, a_{2024}$ be an Arithmetic Progression such that

$a_1 + (a_5 + a_{10} + a_{15} + \dots + a_{2020}) + a_{2024} = 2233$. Then $a_1 + a_2 + a_3 + \dots + a_{2024}$ is equal to

[JEE Mains-2025]

[Ans. : 11132]

COMMON TERMS



DIBY-25

The 8th common term of the series

$$S_1 = 3 + 7 + 11 + 15 + 19 + \dots$$

$$S_2 = 1 + 6 + 11 + 16 + 21 + \dots \text{ is}$$

[JEE Mains-2023]

[Ans. : 151]

DIBY-26

Let 3, 6, 9, 12, upto 78 terms and 5, 9, 13, 17, upto 59 terms be two series. Then, the sum of terms common to both the series is equal to _____.

[JEE Mains-2022]

[Ans. : 2223]

DIBY-27

The number of terms common to the two A.P.s 3, 7, 11, ..., 407 and 2, 9, 16,, 709 is

[JEE Mains-2020]

DIBY-28

[Ans. : 17]

Let 3, 7, 11, 15, ..., 403 and 2, 5, 8, 11, ..., 404 be two arithmetic progressions. Then the sum, of the common terms in them, is equal to

[JEE Mains-2024]

[Ans. : 6699]

GEOMETRIC PROGRESSION



DIBY-29

Let $a_1, a_2, a_3, \dots$ be a G. P. of increasing positive numbers. If $a_3 a_5 = 729$ and $a_2 + a_4 = \frac{11T}{4}$, then $24(a_1 + a_2 + a_3)$ is equal to **[JEE Mains-2025]**

(A) 131 (B) 130 (C) 129 (D) 128

[Ans. : c]

DIBY-30

Let $a_1, a_2, a_3, \dots$ be a G.P. of increasing positive terms. If $a_1 a_5 = 28$ and $a_2 + a_4 = 29$, then a_6 is equal to: **[JEE Mains-2025]**

(A) 812 (B) 526 (C) 784 (D) 628

[Ans. : c]

DIBY-31

Let a and b be two distinct positive real numbers. Let 11th term of a GP, whose first term is a and third term is b , is equal to p^{th} term of another GP, whose first term is a and fifth term is b . Then p is equal to **[JEE Mains-2024]**

(A) 20 (B) 25 (C) 21 (D) 24

[Ans. : c]

GEOMETRIC PROGRESSION



DIBY-32

Let $a_1, a_2, a_3, \dots$ be a G.P. of increasing positive numbers. Let the sum of its 6th and 8th terms be 2 and the product of its 3rd and 5th terms be $\frac{1}{9}$. Then $6(a_2 + a_4)(a_4 + a_6)$ is equal to **[JEE Mains-2023]**

- (A) $2\sqrt{2}$ (B) 2 (C) $3\sqrt{3}$ (D) 3

[Ans. : D]

DIBY-33

In an increasing geometric progression of positive terms, the sum of the second and sixth terms is $\frac{70}{3}$ and the product of the third and fifth terms is 49. Then the sum of the 4th, 6th and 8th terms is: **[JEE Mains-2024]**

- (A) 96 (B) 78 (C) 91 (D) 84

[Ans. : C]

DIBY-34

If the sum of an infinite GP a, ar, ar^2, ar^3 , the squares of its each term is 150, then the sum of ar^2, ar^4, ar^6 , is: **[JEE Mains-2021]**

- (A) $\frac{1}{2}$ (B) $\frac{25}{2}$ (C) $\frac{5}{2}$ (D) $\frac{9}{2}$

[Ans. : A]

DIBY-35

Three numbers are in an increasing geometric progression with common ratio r . If the middle number is doubled, then the new numbers are in an arithmetic progression with common difference d . If the fourth term of GP is $3r^2$, then $r^2 - d$ is equal to: **[JEE Mains-2021]**

- (A) $7 + \sqrt{3}$ (B) $7 - \sqrt{3}$ (C) $7 + 3\sqrt{3}$ (D) $7 - 7\sqrt{3}$

[Ans. : A]

DIBY-36

If the sum of the second, third and fourth terms of a positive term. G.P. is 3 and the sum of its sixth, seventh and eighth terms is 243, then the sum of the first 50 terms of this G.P. is: **[JEE Mains-2020]**

(A) $\frac{2}{13}(3^{50} - 1)$

(B) $\frac{1}{13}(3^{50} - 1)$

(C) $\frac{1}{26}(3^{49} - 1)$

(D) $\frac{1}{26}(3^{50} - 1)$

[Ans. : D]

DIBY-37

The greatest positive integer k , for which $49^k + 1$ is a factor of the sum $49^{125} + 49^{124} + \dots + 49^2 + 49 + 1$, is **[JEE Mains-2020]**

(A) 32

(B) 63

(C) 65

(D) 60

[Ans. : B]

DIBY-38

If the sum of the second, fourth and sixth terms of a G.P. of positive terms is 21 and the sum of its eighth, tenth and twelfth terms is 15309, then the sum of its first nine terms is: **[JEE Mains-2025]**

(A) 760

(B) 755

(C) 750

(D) 757

[Ans. : D]

DIBY-39

If $x = \sum_{n=0}^{\infty} (-1)^n \tan^{2n} \theta$ and $y = \sum_{n=0}^{\infty} \cos^{2n} \theta$, for $0 < \theta < \frac{\pi}{4}$, then:

[JEE Mains-2020]

(A) $y(1 - x) = 1$

(B) $y(1 + x) = 1$

(C) $x(1 + y) = 1$

(D) $x(1 - y) = 1$

[Ans. : A]

DIBY-40

Let α and β be the roots of $x^2 - 3x + p = 0$ and γ and δ be the roots of $x^2 - 6x + q = 0$. If $\alpha, \beta, \gamma, \delta$ form a geometric progression. Then ratio $(2q + p) : (2q - p)$ is:

[JEE Mains-2020]

(A) 3:1

(B) 5:3

(C) 9:7

(D) 33:31

[Ans. : C]



DIBY-41

If three distinct numbers a, b, c are in G.P. and the equations $ax^2 + 2bx + c = 0$ and $dx^2 + 2ex + f = 0$ have a common root, then which one of the following statements is correct? **[JEE Mains-2019]**

(A) d, e, f are in A.P.

(B) $\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in G.P.

(C) $\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in A.P.

(D) d, e, f are in G.P.

[Ans. : c]

DIBY-42

Find the sum of the following series

(i) $5 + 55 + 555 + \dots$ to n terms

(ii) $0.3 + 0.33 + 0.333 + \dots$ to n terms

Ans. (i) $\frac{5}{9} \left[10 \left(\frac{10^n - 1}{9} \right) n \right]$ (ii) $\frac{1}{3} \left[n - \frac{1}{9} \left(1 - \frac{1}{10^n} \right) \right]$

DIBY-43

If $x = 1 + a + a^2 + a^3 + \dots \infty$ and $y = 1 + b + b^2 + b^3 + \dots \infty$ show that

$1 + ab + a^2b^2 + a^3b^3 + \dots \infty = \frac{xy}{x+y-1}$ where $0 < a < 1$ and $0 < b < 1$.



DIBY-44

For the two positive number a, b if a, b and $\frac{1}{18}$ are in a geometric progression, while $\frac{1}{a}, 10$ and $\frac{1}{b}$ are in an arithmetic progression, then $16a + 12b$ is equal to **[JEE Mains-2023]**

[Ans. 3]

DIBY-45

If the first 3 consecutive terms of a geometrical progression are the roots of the equation $2x^3 - 19x^2 + 57x - 54 = 0$ find the sum to infinite number of terms of G.P.

[Ans. $\frac{27}{2}$]

DIBY-46

Let x_1, x_2, x_3, x_4 be in a geometric progression. If 2, 7, 9, 5 are subtracted respectively from x_1, x_2, x_3, x_4 then the resulting numbers are in an arithmetic progression. Then the value of $\frac{1}{24}(x_1 x_2 x_3 x_4)$ is: **[JEE Mains-2025]**

(A) 72

(B) 18

(C) 36

(D) 216

[Ans.]

DIBY-47

If p arithmetic mean are inserted between 5 and 41 so that ratio $\frac{A_3}{A_{p-1}} = \frac{2}{5}$, then find the value of p .

[Ans. 11]

DIBY-48

If $a_1 a_2 a_3 \dots a_n$ are n arithmetic means inserted between 7 and 2015 whose sum is 56616, then

(A) n is 56

(B) n is 28

(B) $a_{19} = 2029/3$

(D) $a_{19} = 36543/57$

[Ans. c]

DIBY-49

The sides of a right angled triangle are in arithmetic progression. If the triangle has area 24, then what is the length of its smallest side?

[Ans. 6]

DIBY-50

Let $S = 109 + \frac{108}{5} + \frac{107}{5^2} + \dots + \frac{2}{5^{107}} + \frac{1}{5^{108}}$. Then the value of $(16S - (25)^{-54})$ is equal to

[JEE Mains-2023]

[Ans. 2175]

DIBY-51

Suppose $a_1, a_2, 2, a_3, a_4$ be in an arithmetico-geometric progression. If the common ratio of the corresponding geometric progression is 2 and the sum of all 5 terms of the arithmetico-geometric progression is $\frac{49}{2}$, then a_4 is equal to _____. **[JEE Mains-2023]**

[Ans. 16]

DIBY-52

If $7 = 5 + \frac{1}{7}(5 + \alpha) + \frac{1}{7^2}(5 + 2\alpha) + \frac{1}{7^3}(5 + 3\alpha) + \dots \dots \infty$, then the value of α is : **[JEE Mains-2025]**

(A) $\frac{6}{7}$

(B) $\frac{1}{7}$

(C) 6

(D) 1

[Ans. C]

DIBY-53

Let $S = 2 + \frac{6}{7} + \frac{12}{7^2} + \frac{20}{7^3} + \frac{30}{7^4} + \dots$. Then $4S$ is equal to

[JEE Mains-2022]

- (A) $\left(\frac{7}{3}\right)^2$ (B) $\frac{7^3}{3^2}$ (C) $\left(\frac{7}{3}\right)^3$ (D) $\frac{7^2}{3^3}$

[Ans. c]

DIBY-54

If $\frac{6}{3^{12}} + \frac{10}{3^{11}} + \frac{20}{3^{10}} + \frac{40}{3^9} + \dots + \frac{10240}{3} = 2^n \cdot m$, where m is odd, then mn is equal to

[JEE mains-2022]

[Ans. 12]

DIBY-55

If $S = \frac{7}{5} + \frac{9}{5^2} + \frac{13}{5^3} + \frac{19}{5^4} + \dots$, then $160S$ is equal to

[JEE Mains-2021]

[Ans. 305]

DIBY-56

If $S(x) = (1+x) + 2(1+x)^2 + 3(1+x)^3 + \dots + 60(1+x)^{60}$, $x \neq 0$, and $(60)^2 S(60) = a(b)^b + b$, where $a, b \in N$, then $(a+b)$ equal to

[JEE Mains-2024]

[Ans. 3660]



DIBY-57

Let a and b be positive real numbers. If a, A_1, A_2, b are in arithmetic progression, a, G_1, G_2, b are in geometric progression and a, H_1, H_2, b are in harmonic progression, show that

$$\frac{G_1 G_2}{H_1 H_2} = \frac{A_1 + A_2}{H_1 + H_2} = \frac{(2a+b)(a+2b)}{9ab}.$$

DIBY-58

Find the sum $2017 + \frac{1}{4} \left(2016 + \frac{1}{4} \left(2015 + \dots + \frac{1}{4} \left(2 + \frac{1}{4} (1) \right) \dots \right) \right)$

$$[\text{Ans. } S = \frac{4}{3} (2017) - \frac{4}{9} \left(1 - \frac{1}{4^{2017}} \right)]$$

DIBY-59

Find the sum $1 + 2 \left(1 + \frac{1}{50} \right) + 3 \left(1 + \frac{1}{50} \right)^2 + \dots$ 50 terms .

$$[\text{Ans. :2500}]$$

DIBY-60

Find the sum $\frac{1^2}{2} + \frac{3^2}{2^2} + \frac{5^2}{2^3} + \frac{7^2}{2^4} + \dots \infty$

[Ans. 17]

DIBY-61

If the sum of the first 40 terms of the series, $3 + 4 + 8 + 9 + 13 + 14 + 18 + 19 + \dots$ is $(102)m$, then m is equal to:

[JEE Mains-2020]

(A) 25

(B) 20

(C) 10

(D) 5

[Ans. (B)]

DIBY-62

If the range of $f(\theta) = \frac{\sin^4 \theta + 3\cos^2 \theta}{\sin^4 \theta + \cos^2 \theta}$, $\theta \in \mathbb{R}$ is $[\alpha, \beta]$, then the sum of the infinite G.P., whose first term is 64 and the common ratio is $\frac{\alpha}{\beta}$, is equal to

[JEE Mains-2024]

[Ans. 96]

DIBY-63

Let $a, b, c > 1$, a^3, b^3 and c^3 be in A.P., and $\log_a b, \log_c a$ and $\log_b c$ be in G.P. If the sum of first 20 terms of an A.P., whose first term is $\frac{a+4b+c}{3}$ and the common difference is $\frac{a-8b+c}{10}$ is -444 , then abc is equal to

[JEE Mains-2023]

(A) 343

(B) 216

(C) $\frac{343}{8}$

(D) $\frac{125}{8}$

[Ans. (B)]



PROPERTIES OF SIGMA

DIBY-64

The sum $\sum_{k=1}^{20} (1 + 2 + 3 + \dots + k)$ is

[JEE Mains-2020]

[Ans. 1540]

DIBY-65

Let S_n be the sum to n -terms of an arithmetic progression 3, 7, 11,

If $40 < \left(\frac{6}{n(n+1)} \sum_{k=1}^n S_k \right) < 42$, then n equals

[JEE Mains-2024]

[Ans. 9]

DIBY-66

Let $A = \sum_{i=1}^{10} \sum_{j=1}^{10} \min\{i, j\}$ and $B = \sum_{i=1}^{10} \sum_{j=1}^{10} \max\{i, j\}$. Then $A + B$ is equal to

[JEE Mains-2022]

[Ans. 1100]

DIBY-67

Let $S_K = \frac{1+2+\dots+K}{K}$ and $\sum_{j=1}^n S_j^2 = \frac{n}{A}(Bn^2 + Cn + D)$ where $A, B, C, D \in \mathbb{N}$ and A has least value.

Then

[JEE Mains-2023]

(A) $A + C + D$ is not divisible by B

(B) $A + B = 5(D - C)$

(C) $A + B + D$ is divisible by 5

(D) $A + B$ is divisible by D

[Ans. (D)]

DIBY-68

If $\frac{1^3+2^3+3^3+\dots \text{ upto } n \text{ terms}}{1 \cdot 3+2 \cdot 5+3 \cdot 7+\dots \text{ upto } n \text{ terms}} = \frac{9}{5}$, then the value of n is

[JEE Mains-2023]

[Ans. 5]

DIBY-69

The sum, $\sum_{n=1}^7 \frac{n(n+1)(2n+1)}{4}$ is equal to

[JEE Mains-2020]

[Ans. 504]

DIBY-70

If $\gcd(m, n) = 1$ and $1^2 - 2^2 + 3^2 - 4^2 + \dots + (2021)^2 - (2022)^2 + (2023)^2 = 1012m^2n$, then $m^2 - n^2$ is equal to

[JEE Mains-2023]

(A) 200

(B) 240

(C) 220

(D) 180

[Ans. (B)]



DIBY-71

If $1 + (1 - 2^2 \cdot 1) + (1 - 4^2 \cdot 3) + (1 - 6^2 \cdot 5) + \dots + (1 - 20^2 \cdot 19) = \alpha - 220\beta$, then an ordered pair (α, β) is equal to: **[JEE Mains-2020]**

- (A) (10, 103) (B) (10, 97) (C) (11, 97) (D) (11, 103)

[Ans. (D)]

DIBY-72

Let $S_k = \frac{1+2+3+\dots+k}{k}$. If $S_1^2 + S_2^2 + \dots + S_{10}^2 = \frac{5}{12}A$, then A equal to: **[JEE Mains-2019]**

- (A) 283 (B) 301 (C) 303 (D) 156

[Ans. (c)]

DIBY-73

The sum of the series $1 + 2 \times 3 + 3 \times 5 + 4 \times 7 + \dots$ upto 11th term is: **[JEE Mins-2019]**

- (A) 915 (B) 946 (C) 945 (D) 916

[Ans. (B)]

DIBY-74

If the sum of the first 15 terms of the series $\left(\frac{3}{4}\right)^3 + \left(1\frac{1}{2}\right)^3 + \left(2\frac{1}{4}\right)^3 + 3^3 + \left(3\frac{3}{4}\right)^3 + \dots$ is equal to $225k$, then k is equal to: **[JEE Mains-2019]**

(A) 108

(B) 27

(C) 54

(D) 9

[Ans. (c)]

WRITING T_n FOR SERIES WITH COMMON DIFFERENCE IN AP OR GP



DIBY-75

$S_n = 2 + 5 + 14 + 41 + 122 + \dots$ Upto n -terms.

[Ans. $\frac{3^{n+1} + 2n - 3}{4}$]

DIBY-76

The sum $1 + 3 + 11 + 25 + 45 + 71 + \dots$ upto 20 terms, is equal to

- (A) 7240 (B) 7130 (C) 6982 (D) 8124

[JEE Mains-2025]

[Ans. (A)]

DIBY-77

If $S_n = 4 + 11 + 21 + 34 + 50 + \dots$ to n terms, then $\frac{1}{60}(S_{29} - S_9)$

- (A) 226 (B) 220 (C) 223 (D) 227

[JEE Mains-2023]

[Ans. (c)]

TELESCOPIC METHOD OF DIFFERENCE



DIBY-78

$\frac{1}{3^2-1} + \frac{1}{5^2-1} + \frac{1}{7^2-1} + \dots + \frac{1}{(201)^2-1}$ is equal to

[JEE Mains-2021]

(A) $\frac{101}{404}$

(B) $\frac{25}{101}$

(C) $\frac{101}{408}$

(D) $\frac{99}{400}$

[Ans. (B)]

DIBY-79

If the sum of series $\frac{1}{1 \cdot (1+d)} + \frac{1}{(1+d)(1+2d)} + \dots + \frac{1}{(1+9d)(1+10d)}$ is equal to 5, then $50d$ is equal to:

[JEE Mains-2024]

(A) 20

(B) 5

(C) 15

(D) 10

[Ans. (A)]

DIBY-80

The sum $\sum_{n=1}^{21} \frac{3}{(4n-1)(4n+3)}$ is equal to

[JEE Mains-2022]

(A) $\frac{7}{87}$

(B) $\frac{7}{29}$

(C) $\frac{14}{87}$

(D) $\frac{21}{29}$

[Ans. (B)]

DIBY-81

The sum of 10 terms of the series $\frac{3}{1^2 \times 2^2} + \frac{5}{2^2 \times 3^2} + \frac{7}{3^2 \times 4^2} + \dots$ is

(A) $\frac{143}{144}$

(B) 1

(C) $\frac{99}{100}$

(D) $\frac{120}{121}$

[JEE Mains-2021]

[Ans. (D)]

DIBY-82

If $\frac{1}{2 \times 3 \times 4} + \frac{1}{3 \times 4 \times 5} + \frac{1}{4 \times 5 \times 6} + \dots + \frac{1}{100 \times 101 \times 102} = \frac{k}{101}$, then $34k$ is equal to

[JEE Mains-2022]

[Ans. 286]

TELESCOPIC METHOD OF DIFFERENCE



DIBY-83

If $\sum_{k=1}^{10} \frac{k}{k^4+k^2+1} = \frac{m}{n}$, where m and n are co-prime, then $m + n$ is equal to **[JEE Mains-2022]**
[Ans. 166]

DIBY-84

The sum to 10 terms of the series $\frac{1}{1+1^2+1^4} + \frac{2}{1+2^2+2^4} + \frac{3}{1+3^2+3^4} + \dots$ is **[JEE Mains-2023]**
(A) $\frac{59}{111}$ (B) $\frac{55}{111}$ (C) $\frac{56}{111}$ (D) $\frac{58}{111}$ [Ans. (B)]

DIBY-85

If $a_n = \frac{-2}{4n^2-16n+15}$, then $a_1 + a_2 + \dots + a_{25}$ is equal to: **[JEE Mains-2023]**
(A) $\frac{51}{144}$ (B) $\frac{49}{138}$ (C) $\frac{50}{141}$ (D) $\frac{52}{147}$ [Ans. (c)]

TELESCOPIC METHOD OF DIFFERENCE



DIBY-86

The sum of the series $\frac{1}{1-3 \cdot 1^2+1^4} + \frac{2}{1-3 \cdot 2^2+2^4} + \frac{3}{1-3 \cdot 3^2+3^4} + \dots$ up to 10 terms is **[JEE Mains-2024]**

- (A) $\frac{45}{109}$ (B) $-\frac{45}{109}$ (C) $\frac{55}{109}$ (D) $-\frac{55}{109}$

[Ans.: (D)]

DIBY-87

If the sum of the first 20 terms of the series $\frac{4.1}{4+3 \cdot 1^2+1^4} + \frac{4.2}{4+3 \cdot 2^2+2^4} + \frac{4.3}{4+3 \cdot 3^2+3^4} + \frac{4.4}{4+3 \cdot 4^2+4^4} + \dots$ is $\frac{m}{n}$, where m and n are coprime, then $m + n$ is equal to

- (A) 423 (B) 420 (C) 421 (D) 422

[Ans.: (C)]

DIBY-88

The sum $\sum_{k=1}^n \frac{1}{(k+1)\sqrt{k}+k\sqrt{k+1}}$ is equal to

[Ans.: $\frac{\sqrt{n+1}-1}{\sqrt{n+1}}$]

TELESCOPIC METHOD OF DIFFERENCE



DIBY-89

Find sum of n -terms of the series: $S = \frac{1}{2 \cdot 3} \times 2 + \frac{2}{3 \cdot 4} \times 2^2 + \frac{3}{4 \cdot 5} \times 2^3 + \dots$

$$[\text{Ans.: } S = \frac{2^{n+1}}{n+2} - 1]$$

DIBY-90

$S = \frac{5}{1 \cdot 2} \times \frac{1}{3} + \frac{7}{2 \cdot 3} \times \frac{1}{3^2} + \frac{9}{3 \cdot 4} \times \frac{1}{3^3} + \frac{11}{4 \cdot 5} \times \frac{1}{3^4} + \dots$ upto n -terms

$$[\text{Ans.: } S = 1 - \frac{1}{3^n(n+1)}]$$

TELESCOPIC METHOD OF DIFFERENCE



DIBY-91

Let for $n = 1, 2, \dots, 50$ be the sum of the infinite geometric progression whose first term is n^2 and whose common ratio is $\frac{1}{(n+1)^2}$. Then the value of $\frac{1}{26} + \sum_{n=1}^{50} \left(S_n + \frac{2}{n+1} - n - 1 \right)$ is equal to

[JEE Mains-2022]

[Ans. 41651]

DIBY-92

Let $S_n = \frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \frac{1}{20} + \dots$ upto n terms. If the sum of the first six terms of an A.P. with first term $-p$ and common difference p is $\sqrt{2026S_{2025}}$, then the absolute difference between 20th and 15th terms of the A.P. is

[JEE Mains-2025]

(A) 20

(B) 45

(C) 25

(D) 90

[Ans. (c)]

TELESCOPIC METHOD OF DIFFERENCE



DIBY-93

Find the sum $\sum_{n=1}^{\infty} \frac{6^n}{(3^n - 2^n)(3^{n+1} - 2^{n+1})}$

[Ans. (2)]

DIBY-94

Find the sum of the series $\frac{1}{3^2+1} + \frac{1}{4^2+2} + \frac{1}{5^2+3} + \frac{1}{6^2+4} + \dots \infty$.

[Ans. (13/36)]

DIBY-95

Find the sum $\sum_{n=2}^{\infty} \frac{3n^2+1}{(n^2-1)^3}$

[Ans. (09/16)]

DIBY-96

Let $S = \frac{\sqrt{1}}{1+\sqrt{1}+\sqrt{2}} + \frac{\sqrt{2}}{1+\sqrt{2}+\sqrt{3}} + \frac{\sqrt{3}}{1+\sqrt{3}+\sqrt{4}} + \dots + \frac{\sqrt{n}}{1+\sqrt{n}+\sqrt{n+1}} = 10$. Then find the value of n .

[Ans. (24)]

INEQUALITY

DIBY-97

Minimum value of sum of real numbers $a^{-5}, a^{-4}, 3a^{-3}, 1, a^8$ & a^{10} (where $a > 0$)
[IIT-JEE-2011]

[Ans. 8]

DIBY-98

If $a > 0, b > 0, c > 0$, then show that: $(a + b + c) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) \geq 9$ [JEE-1984]

DIBY-99

For positive real number a, b & c show that $((1 + a)(1 + b)(1 + c))^7 > 7^7 a^4 b^4 c^4$ [JEE-2004]

DIBY-100

Let $x, y > 0$. If $x^3 y^2 = 215$, then the least value of $3x + 2y$ is [JEE Mains-2022]
(A) 30 (B) 32 (C) 36 (D) 40

[Ans. (D)]

INEQUALITY

DIBY-101

The minimum value of $f(x) = a^{a^x} + a^{1-x^x}$, where $a, x \in R$ and $a > 0$, is equal to

[JEE Mains-2021]

- (A) $a + \frac{1}{a}$ (B) $2a$ (C) $a + 1$ (D) $2\sqrt{a}$

[Ans.]

DIBY-102

Let a, b, c and d be positive real numbers such that $a + b + c + d = 11$. If the maximum value of $a^5 b^3 c^2 d$ is 3750β , then the value of β is

[JEE Mains-2023]

- (A) 110 (B) 108 (C) 90 (D) 55

[Ans.]

DIBY-103

If the minimum value of $f(x) = \frac{5x^2}{2} + \frac{\alpha}{x^5}$, $x > 0$ is 14, then the value of α is equal to

[JEE Mains-2022]

- (A) 32 (B) 64 (C) 128 (D) 256

[Ans. (c)]

EXPONENTIAL & LOG SERIES



DIBY-104

Simplify : $S = \frac{2}{1!} + \frac{4}{3!} + \frac{6}{5!} + \frac{8}{7!} + \dots \infty$

[Ans. (e)]

DIBY-105

$$S = 1 + \frac{1+2}{2!} + \frac{1+2+2^2}{3!} + \frac{1+2+2^2+2^3}{4!} + \dots \infty$$

[Ans. ($e^2 - e$)]

DIBY-106

The sum $1 + \frac{1+3}{2!} + \frac{1+3+5}{3!} + \frac{1+3+5+7}{4!} + \dots$ upto ∞ terms, is equal to

(A) $6e$

(B) $4e$

(C) $3e$

(D) $2e$

[JEE Mains-2025]

[Ans. (D)]

EXPONENTIAL & LOG SERIES



DIBY-107

The sum of the series $\sum_{n=1}^{\infty} \frac{n^2+6n+10}{(2n+1)!}$ is equal to:

[JEE Mains-2021]

(A) $\frac{41}{8}e + \frac{19}{8}e^{-1} - 10$

(B) $\frac{41}{8}e - \frac{19}{8}e^{-1} - 10$

(C) $\frac{41}{8}e + \frac{19}{8}e^{-1} + 10$

(D) $-\frac{41}{8}e + \frac{19}{8}e^{-1} - 10$

[Ans. : B]

DIBY-108

If $0 < x < 1$ and $y = \frac{1}{2}x^2 + \frac{2}{3}x^3 + \frac{3}{4}x^4 + \dots$, then the value of e^{1+y} at $x = \frac{1}{2}$ is:

[JEE Mains-2021]

(A) $2e^2$

(B) $\frac{1}{2}e^2$

(C) $2e$

(D) $\frac{1}{2}\sqrt{e}$

[Ans. : B]

Let $S_n = 1 \cdot (n-1) + 2 \cdot (n-2) + 3 \cdot (n-3) + \dots + (n-1) \cdot 1, n \geq 4$.

The sum $\sum_{n=4}^{\infty} \left(\frac{2S_n}{n!} - \frac{1}{(n-2)!} \right)$ is equal to

- (A) $\frac{e-1}{3}$
 (B) $\frac{e-2}{6}$
 (C) $\frac{e}{3}$
 (D) $\frac{e}{6}$

[Ans. : A]

QUESTION [JEE Mains-2024]

DIBY-110



If $1 + \frac{\sqrt{3}-\sqrt{2}}{2\sqrt{3}} + \frac{5-2\sqrt{6}}{18} + \frac{9\sqrt{3}-11\sqrt{2}}{36\sqrt{3}} + \frac{49-20\sqrt{6}}{180} + \dots$ upto $\infty = 2 + \left(\sqrt{\frac{b}{a}} + 1\right) \log_e \left(\frac{a}{b}\right)$,

where a and b are integers with $\gcd(a, b) = 1$, then $11a + 18b$ is equal to

[Ans. : 76]

CHALLENGER QUESTION

DIBY-111

Let $a = \frac{x}{1+x^2} + \frac{1}{3} \left(\frac{x}{1+x^2}\right)^3 + \frac{1}{5} \left(\frac{x}{1+x^2}\right)^5 + \dots$ and $b = x - \frac{2}{3}x^3 + \frac{1}{5}x^5 + \frac{1}{7}x^7 - \frac{2}{9}x^9 \dots$,

then

[Ans. : $a = b$]



*It is never about **end result**
It's always about the **Journey***

THANKYOU

#future **IITians**